

Archimedean first-slab positivity for shifted zeta canonical systems

An off-center Weil generator and a radial energy identity

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This manuscript reports computer-assisted mathematics and was prepared with substantial language-model assistance. The author is responsible for its content. The analytic inputs, the interval-arithmetic certificate, and the independent audit are described in Sections 7 and 8 and Appendix B; the present verification status of every component, including the items that have not yet received line-by-line human verification, is recorded in Section 13 and Appendix C. The preprint is deposited to fix a citable version of the argument while that verification proceeds.

Abstract

For $\omega > 0$, Suzuki's shifted scattering function $\Theta_\omega(z) = \xi(\frac{1}{2} - \omega - iz)/\xi(\frac{1}{2} + \omega - iz)$ admits a prime-gamma Hankel kernel. By results of Suzuki, Θ_ω is a meromorphic inner function in the upper half-plane exactly when the associated Hankel transform is an isometry, and innerness for every $\omega > \omega_0$ is equivalent to the zero-free half-plane $\operatorname{Re} s > \frac{1}{2} + \omega_0$; the Riemann hypothesis is the case of all $\omega_0 > 0$. Contractivity of every finite section of the Hankel operator is therefore a necessary condition for these open assertions when $0 < \omega < \frac{1}{2}$.

We isolate the first logarithmic radial slab, before the first prime layer appears. On a slab of length $L \leq \log 2$ the arithmetic Hankel operator reduces exactly to a reflected finite-horizon Volterra convolution $H_{\omega,L} = V_{\omega,L} R_L$ with completed-gamma transfer $K_\omega(p) = \Gamma_\infty(\frac{1}{2} + p - \omega)/\Gamma_\infty(\frac{1}{2} + p + \omega)$. Differentiating this transfer in the shift produces a self-adjoint generator $Q_{\omega,L}$, an off-center prime-free Weil form, with the exact decomposition $Q_{\omega,L} = Q_{0,L} + C_{\omega,L}$, where $Q_{0,L}$ is the central compact-window Weil form and $C_{\omega,L}$ is the compact integral operator with kernel $c_\omega(x,y) = (\cosh(\omega|x-y|) - 1)(e^{|x-y|/2} - e^{-5|x-y|/2}/(1 - e^{-2|x-y|}))$. The apparent singularity at $\omega = \frac{1}{2}$ cancels in this representation and all generators share one logarithmic form domain. An Arb-certified Legendre calculation, closed by analytic Taylor and infinite-tail bounds, proves $Q_{\omega,L} \succ 0$ for $0 \leq \omega \leq \frac{1}{2}$ and $0 < L \leq \log 2$, with an explicit coercivity constant 6.1159×10^{-4} on the whole form domain. The certificate uses 256 shift boxes in each parity sector and sixteen Legendre modes per sector; a clean-room reimplementaion reproduces all 512 boxes.

We then prove a strong-operator energy identity expressing the defect $I - V_{\omega,L}^* V_{\omega,L}$ as $2 \int_0^\omega V_{s,L}^* Q_{s,L} V_{s,L} ds$. Strict generator positivity, the strong limit $V_{s,L} \rightarrow I$ as $s \downarrow 0$, and compactness of $V_{\omega,L}$ give $\|V_{\omega,L}\| < 1$ and $I \pm H_{\omega,L} \succ 0$ throughout the first slab for $0 < \omega \leq \frac{1}{2}$. This complements Suzuki's unconditional strict contractivity of all finite sections for $\omega > \frac{1}{2}$ and settles the critical shift interval on the prime-free slab. The energy identity also exhibits an explicit shift-radial Gram factor with local density $Q_{s,L}$. The result is prime-free and finite-depth: it proves neither the Riemann hypothesis nor a new zero-free region. Its content is an independently positive archimedean base slab for the shifted-zeta canonical-system program.

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1 Introduction

1.1 Shifted zeta positivity

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) \quad (1.1)$$

be the completed Riemann xi-function and, following Suzuki [1], put for $\omega > 0$

$$\Theta_\omega(z) = \frac{\xi(\frac{1}{2} - \omega - iz)}{\xi(\frac{1}{2} + \omega - iz)}. \quad (1.2)$$

The functional equation $\xi(s) = \xi(1-s)$ gives, for real u ,

$$\Theta_\omega(u) = \frac{\xi(\frac{1}{2} + \omega + iu)}{\xi(\frac{1}{2} + \omega - iu)}, \quad |\Theta_\omega(u)| = 1. \quad (1.3)$$

This boundary unitarity is unconditional. The nontrivial condition is *meromorphic innerness* in the upper half-plane $\mathbb{C}_+ = \{\text{Im } z > 0\}$: in the terminology of [1], a bounded analytic function on $\mathbb{C}_+$ with unimodular boundary values almost everywhere which extends meromorphically to $\mathbb{C}$.

We pin the relation to zero-free regions to the statements actually proved in [1], since it carries the significance claim of this paper. Suzuki proves [1, Prop. 1.2] that for $\omega_0 > 0$ the following are equivalent: (i) $\zeta(s) \neq 0$ for $\text{Re } s > \frac{1}{2} + \omega_0$; (ii) Θ_ω is a meromorphic inner function in $\mathbb{C}_+$ for every $\omega > \omega_0$. In particular the Riemann hypothesis is equivalent to meromorphic innerness of Θ_ω for every $\omega > 0$, and for $0 < \omega_0 < \frac{1}{2}$ the half-plane statement (i) is an open quasi-RH assertion. The equivalence is stated for the whole family $\omega > \omega_0$; for a single shift, innerness of Θ_ω excludes every zero ρ of ξ with $\text{Re } \rho > \frac{1}{2} + \omega$ whose pole $z = i(\rho - \frac{1}{2} - \omega)$ of (1.2) is not cancelled by a zero of the numerator, and the passage to all $\omega > \omega_0$ removes the cancellation cases. Suzuki further shows [1, Thm. 2.2] that Θ_ω is meromorphic inner if and only if the Hankel operator H_ω of Section 2 maps $L^2((1, \infty), dx)$ into $L^2((0, \infty), dx)$ —equivalently, if and only if on $L^2((1, \infty), dx)$ the multiplier action of Θ_ω coincides with H_ω —and that in this case H_ω is an isometry of $L^2((0, \infty), dx)$ [1, Lemma 4.1]. Every compression of an isometry is a contraction. Hence contractivity of every finite section of H_ω is a necessary condition for innerness, and thus, for $0 < \omega < \frac{1}{2}$, for the open zero-free statements; compare the classical positivity formulations of Weil and Lagarias [7]. Theorem A verifies this necessary condition unconditionally on the entire first slab and throughout the critical shift interval $0 < \omega \leq \frac{1}{2}$. For $\omega > \frac{1}{2}$ the zero-free statement follows from the classical critical strip, and Suzuki [1, Lemma 4.4] proves the strict finite-section contractivity there; see Section 9.3. The genuinely RH-sensitive shift range is the one treated here.

The global problem is arithmetic. Suzuki’s Hankel kernel contains translated copies of one archimedean impulse at logarithmic depths $\log n$, weighted by Euler-product coefficients. Proving compatible contractivity at every depth for shifts accumulating at zero is an RH-level problem.

The present paper addresses a local but independently specified part of that system. Before the first threshold $\log 2$, no prime layer is present. The kernel is determined solely by the completed gamma factor, so its positivity can be studied without zeta zeros, a zero-free hypothesis, or an assumed positive spectral measure.

1.2 The first slab and the proof architecture

An earlier approach to the first slab proceeded through the singular limit $\omega \downarrow 0$, compact Mosco convergence to a prime-free Weil form, and direct parameter enclosures for the defect channels. That route remains a useful cross-check (Section 11), but it obscures a simpler monotonicity mechanism.

The key observation is to differentiate the causal transfer in ω . If

$$\partial_\omega V_{\omega,L} = -A_{\omega,L} V_{\omega,L}, \quad (1.4)$$

then the defect $D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}$ satisfies

$$\partial_\omega D_{\omega,L} = V_{\omega,L}^* (A_{\omega,L} + A_{\omega,L}^*) V_{\omega,L}. \quad (1.5)$$

Thus contractivity follows if the self-adjoint shift generator is positive. The generator is not a new unrelated form: it is an off-center deformation of the same compact-window Weil form that appears at $\omega = 0$. More precisely, after the normalization

$$Q_{\omega,L} := \frac{1}{2} (A_{\omega,L} + A_{\omega,L}^*), \quad (1.6)$$

all shift dependence is a small bounded compact kernel. This is the structural reason the full shift interval is certifiable with a modest matrix calculation.

1.3 Main results

Let $R_L f(x) = f(L - x)$ on $L^2(0, L)$. Let $V_{\omega, L}$ be the causal Volterra operator defined in Section 2, and put $H_{\omega, L} = V_{\omega, L} R_L$.

Theorem A (critical-shift first-slab positivity). *For every*

$$0 < \omega \leq \frac{1}{2}, \quad 0 < L \leq \log 2,$$

one has

$$\|V_{\omega, L}\| = \|H_{\omega, L}\| < 1, \quad \text{and therefore} \quad I \pm H_{\omega, L} \succ 0. \quad (1.7)$$

Theorem A follows from Theorems B and C below. Section 9.3 records the relation to Suzuki's complementary range $\omega > \frac{1}{2}$ without using that range in Theorem A. The computer-assisted input is isolated in one theorem.

Theorem B (computer-assisted off-center Weil positivity). *Let $0 \leq \omega \leq \frac{1}{2}$ and $0 < L \leq \log 2$. Then the closed form $Q_{\omega, L}$ with logarithmic form domain $\mathcal{D}_{\log}(I_L)$ (Section 4.4) is strictly positive. More precisely, for every $f \in \mathcal{D}_{\log}(I_L)$,*

$$Q_{\omega, L}[f] \geq c_* \|f\|_2^2, \quad c_* = 6.1159 \times 10^{-4}, \quad (1.8)$$

and c_ may be replaced by 5.2613×10^{-2} on the odd reflection sector. At $L = \log 2$ the primary Arb certificate gives minimum certified Weyl–Schur margins*

$$6.116350109501 \times 10^{-4} \quad \text{and} \quad 5.261670009275 \times 10^{-2}$$

in the even and odd sectors, from which (1.8) follows by Lemma 7.1. A clean-room reconstruction gives $6.116514636999 \times 10^{-4}$ and $5.261670255041 \times 10^{-2}$.

The analytic step from Theorem B to Theorem A is exact.

Theorem C (shift-energy identity). *For $0 < \omega \leq \frac{1}{2}$ and $0 < L \leq \log 2$,*

$$I - V_{\omega, L}^* V_{\omega, L} = 2 \int_0^\omega V_{s, L}^* Q_{s, L} V_{s, L} \, ds, \quad (1.9)$$

where the integral is an improper strong-operator integral. In particular $I - V_{\omega, L}^ V_{\omega, L}$ is positive definite. Moreover $V_{\omega, L}$ is compact, so positive definiteness excludes norm one and gives strict contractivity.*

The energy identity supplies a canonical positive factor. Define

$$(\mathcal{G}_{\omega, L} f)(s) = \sqrt{2} Q_{s, L}^{1/2} V_{s, L} f, \quad 0 < s < \omega. \quad (1.10)$$

Then

$$I - V_{\omega, L}^* V_{\omega, L} = \mathcal{G}_{\omega, L}^* \mathcal{G}_{\omega, L}. \quad (1.11)$$

The shift s is therefore a radial coordinate and $Q_{s, L}$ is the local positive energy density accumulated by the boundary defect.

1.4 Organization of the argument

The generator certificate and the shift-energy identity are the computational and analytic halves of one contractivity theorem. Their combination identifies compact-window Weil positivity as the local dissipation density of the shifted causal system. Sections 2–3 set up Suzuki's system and the completed-gamma transfer; Section 4 derives the off-center generator and its bounded-kernel decomposition; Sections 5–7 contain the certificate; Section 8 proves the energy identity; Sections 9–12 discuss consequences, context, superseded routes, and open problems; Section 13 records the verification status. The fractional Sonine realization, compact Mosco limit, and older direct-transfer certificates are retained as secondary structure and cross-checks.

1.5 What is not claimed

The scope is deliberately limited.

1. The theorem does not prove RH.
2. It gives no new zero-free region. It verifies a necessary finite-section condition for the open shifted zero-free assertions, but a prime-free finite-depth compression cannot determine global innerness.
3. No arithmetic layer $n \geq 2$ is controlled. The first new layer appears exactly at depth $\log 2$.
4. The positive radial factor is a rigorous one-dimensional bulk in the sense of a transfer-system energy identity. It is not claimed to be an ordinary AdS/CFT dual.
5. The proof is computer-assisted. The primary special-function enclosure is formal Arb arithmetic; the independent implementation is a strong audit rather than a second formal special-function interval library.

2 Suzuki's arithmetic Hankel system

2.1 Prime–gamma layer decomposition

Define the arithmetic coefficients [1, eq. (2.1)]

$$c_\omega(n) = n^\omega \sum_{d|n} \frac{\mu(d)}{d^{2\omega}} = n^\omega \prod_{p|n} (1 - p^{-2\omega}), \quad \sum_{n \geq 1} c_\omega(n) n^{-w} = \frac{\zeta(w - \omega)}{\zeta(w + \omega)}. \quad (2.1)$$

Suzuki constructs an explicit archimedean function g_ω supported in $(0, 1)$ and the arithmetic kernel [1, eq. (2.3)]

$$h_\omega(x) = \frac{1}{x} \sum_{n \leq x} c_\omega(n) g_\omega(n/x), \quad x > 1, \quad (2.2)$$

with $h_\omega = 0$ on $(0, 1)$, and proves [1, Prop. 2.1] that $\int_0^\infty h_\omega(x) x^{1/2+iz} dx/x = \Theta_\omega(z)$ for $\text{Im } z > \frac{1}{2} + \omega$. The multiplicative Hankel operator is [1, eq. (2.5)]

$$(\mathbf{H}_\omega f)(x) = \int_0^\infty h_\omega(xy) f(y) dy. \quad (2.3)$$

Introduce logarithmic coordinates

$$k_\omega(t) = e^{t/2} h_\omega(e^t), \quad \kappa_\omega(t) = e^{-t/2} g_\omega(e^{-t}) \mathbf{1}_{t>0}. \quad (2.4)$$

Then (2.2) becomes

$$k_\omega(t) = \sum_{n \geq 1} \frac{c_\omega(n)}{\sqrt{n}} \kappa_\omega(t - \log n). \quad (2.5)$$

Every integer contributes a delayed copy of the same archimedean impulse. A finite sum-coordinate interval of length $L < \log 2$ sees only $n = 1$. At $L = \log 2$ the $n = 2$ contribution lies on a measure-zero boundary and does not enter the quadratic form.

2.2 Finite-volume Hankel operator

On the centered interval $I_L = (-L/2, L/2)$ the first-slab Hankel operator is

$$(H_{\omega,L}^c f)(x) = \int_{I_L} k_\omega(x+y) f(y) dy. \quad (2.6)$$

Because $L \leq \log 2$, the layer formula (2.5) gives $k_\omega = \kappa_\omega$ almost everywhere on the relevant sum-coordinate range. After shifting I_L to $(0, L)$ the same operator is

$$(H_{\omega,L}f)(x) = \int_0^L \kappa_\omega(x+y-L)f(y) dy. \quad (2.7)$$

In Suzuki's multiplicative normalization this is the compression

$$H_{\omega,a} = P_a H_\omega P_a : L^2((0,a), dx) \rightarrow L^2((0,a), dx) \quad (2.8)$$

of [1, eq. (2.6)]. Since $h_\omega(xy) = 0$ for $xy < 1$, the compression acts on $(1/a, a)$, which in logarithmic coordinates is the centered interval of total length $L = 2 \log a$.

Define

$$(V_{\omega,L}f)(x) = \int_0^x \kappa_\omega(x-y)f(y) dy, \quad (R_L f)(x) = f(L-x). \quad (2.9)$$

A change of variables gives the exact identities

$$H_{\omega,L} = V_{\omega,L} R_L = R_L V_{\omega,L}^*, \quad R_L V_{\omega,L} R_L = V_{\omega,L}^*, \quad H_{\omega,L}^2 = V_{\omega,L} V_{\omega,L}^*. \quad (2.10)$$

Hence $H_{\omega,L}$ is self-adjoint and $\|H_{\omega,L}\| = \|V_{\omega,L}\|$. It follows that

$$I \pm H_{\omega,L} \succ 0 \iff \|V_{\omega,L}\| < 1. \quad (2.11)$$

This is the first-slab reflection-positivity criterion.

2.3 Local positivity versus global innerness

If Θ_ω is meromorphic inner, its Hankel transform is an isometry [1, Lemma 4.1] and every finite section is contractive. Conversely, compatible contractivity at all depths recovers the global Hardy-space mapping property in Suzuki's criterion [1, Thm. 2.2]. Positivity at one finite depth is only a local statement. In particular, Theorem A has no direct zero-free consequence.

3 Completed-gamma transfer and fractional structure

3.1 Transfer function

Put

$$\Gamma_\infty(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad (3.1)$$

so that $\xi = \Gamma_\infty \zeta$; this is the factor written $\gamma(s)$ in [1]. Suzuki shows [1, Sect. 3.1] that

$$\int_0^\infty g_\omega(x) x^s \frac{dx}{x} = \frac{\Gamma_\infty(s-\omega)}{\Gamma_\infty(s+\omega)}, \quad \operatorname{Re} s > \max(\omega-2, 1-\omega). \quad (3.2)$$

With $\kappa_\omega(t) = e^{-t/2}g_\omega(e^{-t})$ and $s = p + \frac{1}{2}$ this is the Laplace transform of the first-slab impulse,

$$K_\omega(p) = \int_0^\infty \kappa_\omega(t)e^{-pt} dt = \frac{\Gamma_\infty(\frac{1}{2} + p - \omega)}{\Gamma_\infty(\frac{1}{2} + p + \omega)}. \quad (3.3)$$

For $0 < \omega < \frac{1}{2}$ write

$$\alpha = \frac{1}{2} - \omega, \quad \beta = \frac{1}{2} + \omega. \quad (3.4)$$

Then

$$K_\omega(p) = \pi^\omega \frac{p+\alpha}{p-\alpha} \frac{p-\beta}{p+\beta} \frac{\Gamma((p+\alpha)/2)}{\Gamma((p+\beta)/2)}. \quad (3.5)$$

The pole at $p = \alpha$ obstructs a naive stationary bounded-real argument. It causes no difficulty on a finite horizon and disappears from the bounded generator difference derived in Section 4.

3.2 Explicit local impulse

Let $q(t) = 1 - e^{-2t}$. For $0 < \omega \leq \frac{1}{2}$ the first-slab impulse can be written

$$\kappa_\omega(t) = \pi^\omega \left[\frac{2e^{-\alpha t} q(t)^{\omega-1}}{\Gamma(\omega)} - \frac{4\alpha\omega e^{\alpha t}}{\Gamma(\omega)} B_{q(t)}(\omega, \alpha) - \frac{4\beta\omega e^{-\alpha t} q(t)^\omega}{\Gamma(\omega+1)} \right], \quad (3.6)$$

where $B_q(a, b) = \int_0^q u^{a-1}(1-u)^{b-1} du$ and the middle term is interpreted as zero at $\alpha = 0$. Its Laplace transform is (3.3); by uniqueness of Laplace transforms, (3.6) agrees with $e^{-t/2} g_\omega(e^{-t})$ almost everywhere. Near the diagonal, $\kappa_\omega(t) = O(t^{\omega-1})$ as $t \downarrow 0$, so $\kappa_\omega \in L^1(0, L)$ for every positive shift.

3.3 Diffusive and Sonine factors

The fractional gamma ratio

$$F_\omega(p) = \frac{\Gamma((p+\alpha)/2)}{\Gamma((p+\beta)/2)} \quad (3.7)$$

has impulse

$$a_\omega(t) = \frac{2}{\Gamma(\omega)} e^{-\alpha t} (1-e^{-2t})^{\omega-1} = \sum_{n=0}^{\infty} d_{\omega,n} e^{-\lambda_{\omega,n} t}, \quad \lambda_{\omega,n} = \alpha + 2n, \quad d_{\omega,n} = \frac{2(1-\omega)_n}{\Gamma(\omega) n!} > 0. \quad (3.8)$$

Define the complementary kernel

$$b_\omega(t) = \frac{2}{\Gamma(1-\omega)} e^{-\beta t} (1-e^{-2t})^{-\omega}. \quad (3.9)$$

The gamma recurrence gives the Sonine identity

$$a_\omega * b_\omega = 2e^{-\alpha t}, \quad \text{equivalently} \quad \frac{1}{2}(\partial_t + \alpha) \mathcal{B}_\omega \mathcal{A}_\omega = I \quad (3.10)$$

on the zero-initial-value core. This places the first-slab system in the general Sonine-kernel framework [8]. It also gives the rational collapse

$$F_{1-\omega}(p+1)K_\omega(p) = \frac{2\pi^\omega(p-\beta)}{(p+\beta)(p-\alpha)}. \quad (3.11)$$

These identities motivate a distributed-state realization, but they are not needed for the proof of Theorem A.

4 The off-center Weil generator

4.1 Shift derivative

Set $G(p) = \Gamma_\infty(\frac{1}{2} + p)$ and $\ell(p) = \log G(p)$. Then $K_\omega(p) = G(p-\omega)/G(p+\omega)$ and

$$\partial_\omega K_\omega(p) = -a_\omega(p)K_\omega(p), \quad a_\omega(p) = \ell'(p-\omega) + \ell'(p+\omega). \quad (4.1)$$

In terms of $\alpha = \frac{1}{2} - \omega$ and $\beta = \frac{1}{2} + \omega$, standard gamma and digamma identities [11] give

$$a_\omega(p) = \frac{1}{p+\alpha} + \frac{1}{p-\alpha} + \frac{1}{p+\beta} + \frac{1}{p-\beta} - \log \pi + \frac{1}{2}\psi\left(\frac{p+\alpha}{2}\right) + \frac{1}{2}\psi\left(\frac{p+\beta}{2}\right). \quad (4.2)$$

Let $A_{\omega,L}$ be the maximal finite-horizon causal operator with transfer a_ω . On a smooth zero-initial-value core,

$$\partial_\omega V_{\omega,L} = -A_{\omega,L} V_{\omega,L}. \quad (4.3)$$

Define the self-adjoint generator form

$$Q_{\omega,L} = \frac{1}{2}(A_{\omega,L} + A_{\omega,L}^*). \quad (4.4)$$

At $\omega = 0$ this is the central prime-free compact-window Weil form.

4.2 Fourier-plus-residue representation

Let f be supported on $I_L = (-L/2, L/2)$, with Fourier transform $\widehat{f}(\tau) = \int_{I_L} f(x) e^{-i\tau x} dx$. Define

$$w_\omega(\tau) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{\alpha + i\tau}{2}\right) + \frac{1}{2} \operatorname{Re} \psi\left(\frac{\beta + i\tau}{2}\right) - \log \pi, \quad (4.5)$$

and the evaluation functionals

$$\mathcal{C}_c(f) = \int_{I_L} f(x) \cosh(cx) dx, \quad \mathcal{S}_c(f) = \int_{I_L} f(x) \sinh(cx) dx. \quad (4.6)$$

For $0 \leq \omega < \frac{1}{2}$, contour displacement gives

$$Q_{\omega,L}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} w_\omega(\tau) |\widehat{f}(\tau)|^2 d\tau + |\mathcal{C}_\alpha(f)|^2 + |\mathcal{C}_\beta(f)|^2 - |\mathcal{S}_\alpha(f)|^2 - |\mathcal{S}_\beta(f)|^2. \quad (4.7)$$

Even and odd reflection sectors therefore decouple. The endpoint needs a distributional correction and is recorded explicitly.

Lemma 4.1 (right-endpoint boundary term). *Let $w_{1/2}^{\text{pt}}$ be the pointwise limit of w_ω as $\omega \uparrow \frac{1}{2}$ away from $\tau = 0$, continuously extended at zero. Then*

$$Q_{1/2,L}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} w_{1/2}^{\text{pt}}(\tau) |\widehat{f}(\tau)|^2 d\tau + \frac{1}{2} |\mathcal{C}_0(f)|^2 + |\mathcal{C}_1(f)|^2 - |\mathcal{S}_1(f)|^2. \quad (4.8)$$

Proof. Since $\psi(z) = -1/z + O(1)$ near $z = 0$,

$$\frac{1}{2} \operatorname{Re} \psi\left(\frac{\alpha + i\tau}{2}\right) = -\frac{\alpha}{\alpha^2 + \tau^2} + \frac{1}{2} \operatorname{Re} \psi\left(\frac{i\tau}{2}\right) + o(1),$$

and $\int_{\mathbb{R}} \alpha(\alpha^2 + \tau^2)^{-1} d\tau = \pi$, so $\frac{1}{2} \operatorname{Re} \psi((\alpha + i\tau)/2) \rightarrow \frac{1}{2} \operatorname{Re} \psi(i\tau/2) - \pi\delta_0$ in distributions as $\alpha \downarrow 0$. Since $\widehat{f}(0) = \mathcal{C}_0(f) = \int f$, the delta term subtracts $\frac{1}{2} |\mathcal{C}_0(f)|^2$ from the pointwise-limit formula, while the residue terms $|\mathcal{C}_\alpha|^2 - |\mathcal{S}_\alpha|^2$ tend to $|\mathcal{C}_0|^2$. This is exactly the boundary contribution carried automatically by the bounded-kernel representation of Proposition 4.2 below, whose uniform convergence as $\omega \uparrow \frac{1}{2}$ justifies the limit. $\square$

4.3 Exact bounded perturbation

Define

$$R(t) = e^{t/2} - \frac{e^{-5t/2}}{1 - e^{-2t}}. \quad (4.9)$$

Proposition 4.2 (bounded shift difference). *For $0 \leq \omega \leq \frac{1}{2}$,*

$$Q_{\omega,L} = Q_{0,L} + C_{\omega,L}, \quad (4.10)$$

where $C_{\omega,L}$ is the self-adjoint integral operator on I_L with kernel

$$c_\omega(x, y) = (\cosh(\omega|x - y|) - 1) R(|x - y|). \quad (4.11)$$

Proof. First take $0 \leq \omega < \frac{1}{2}$ and $\operatorname{Re} p$ sufficiently large. The convergent difference must be formed before its summands are separated. From $\psi(z) = -\gamma + \int_0^\infty (e^{-u} - e^{-zu})(1 - e^{-u})^{-1} du$ with $u = 2t$,

$$\frac{1}{2} \psi\left(\frac{p + c}{2}\right) = -\frac{\gamma}{2} + \int_0^\infty \frac{e^{-2t} - e^{-(p+c)t}}{1 - e^{-2t}} dt. \quad (4.12)$$

In the sum with $c = \alpha, \beta$, minus twice the $c = \frac{1}{2}$ term, the constants and the individually divergent e^{-2t} terms cancel. The remaining integrand is integrable at zero and equals

$$-2e^{-pt} \frac{e^{-t/2}}{1 - e^{-2t}} (\cosh(\omega t) - 1). \quad (4.13)$$

Independently, the four rational terms in $a_\omega - a_0$ have inverse Laplace kernel

$$e^{-\alpha t} + e^{\alpha t} + e^{-\beta t} + e^{\beta t} - 4 \cosh(t/2) = 4 \cosh(t/2) (\cosh(\omega t) - 1), \quad (4.14)$$

using $\cosh \alpha t + \cosh \beta t = 2 \cosh(t/2) \cosh(\omega t)$. Adding the two contributions and using

$$2 \cosh(t/2) - \frac{e^{-t/2}}{1 - e^{-2t}} = e^{t/2} - \frac{e^{-5t/2}}{1 - e^{-2t}} = R(t) \quad (4.15)$$

yields the multiplier identity

$$a_\omega(p) - a_0(p) = \int_0^\infty e^{-pt} 2 (\cosh(\omega t) - 1) R(t) dt. \quad (4.16)$$

The causal convolution with this kernel has the kernel $2(\cosh(\omega(x-y)) - 1)R(x-y)\mathbf{1}_{x>y}$; symmetrizing according to (4.4) removes the factor two and gives (4.11). Finally, c_ω converges uniformly on the compact square I_L^2 as $\omega \uparrow \frac{1}{2}$ (the factor $\cosh(\omega t) - 1$ is entire in ω and R is fixed); hence the form identity extends to the endpoint by bounded-operator convergence, consistently with Lemma 4.1. $\square$

4.4 Common logarithmic form domain

As $t \downarrow 0$, $R(t) = -\frac{1}{2t} + O(1)$ while $\cosh(\omega t) - 1 = \frac{1}{2}\omega^2 t^2 + O(t^4)$. Thus $c_\omega(x, y) = O(|x - y|)$ at the diagonal. Hence $C_{\omega, L}$ is Hilbert–Schmidt, and therefore bounded and compact. All $Q_{\omega, L}$ are closed forms on the common domain

$$\mathcal{D}_{\log}(I_L) = \left\{ f \in L^2(I_L) : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{f}(\tau)|^2 d\tau < \infty \right\}. \quad (4.17)$$

This also removes the apparent parameter singularity at $\omega = \frac{1}{2}$.

5 The central prime-free Weil form

5.1 Central form

At $\omega = 0$, $\alpha = \beta = \frac{1}{2}$ and the generator becomes

$$Q_{0, L}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} \left[\operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\tau}{2}\right) - \log \pi \right] |\widehat{f}(\tau)|^2 d\tau + 2|\mathcal{C}_{1/2}(f)|^2 - 2|\mathcal{S}_{1/2}(f)|^2. \quad (5.1)$$

This is the prime-free archimedean Weil form restricted to the compact window I_L . It is diagonal with respect to reflection parity. The logarithmic growth of the digamma multiplier makes $Q_{0, L}$ unbounded above on L^2 , but it is lower bounded and closed on $\mathcal{D}_{\log}$. The residue is positive in the even sector and negative in the odd sector, so both sectors must be checked.

5.2 Finite-frequency reduction

At the maximal window $L = \log 2$ set $T_0 = \frac{1}{2} \log 2$. Let ϕ_n be the normalized Legendre basis on $(-T_0, T_0)$. Its Fourier and pole-evaluation coefficients are

$$\widehat{\phi}_n(\tau) = \sqrt{2T_0(2n+1)} (-i)^n j_n(T_0\tau), \quad p_n = \sqrt{2T_0(2n+1)} i_n(T_0/2), \quad (5.2)$$

where j_n and i_n are the spherical Bessel functions of the first kind and their modified counterparts. Choose the frequency cutoff $\Lambda = 30$ and the high-frequency floor

$$\beta_* = \log \frac{\Lambda}{2\pi} - \frac{1}{\Lambda} = 1.5299869819194765 \dots \quad (5.3)$$

The central matrix is the sum of $\beta_* I$, a compact frequency integral over $[0, 30]$, and the parity-dependent rank-one residue. The complement is controlled by spherical-Bessel majorants.

5.3 Central interval certificate

A sixteen-mode block in each parity sector was evaluated using 80-decimal Arb arithmetic and 48-point Gauss–Legendre quadrature on each of thirty unit frequency panels. All quadrature nodes, weights, digamma values, and Bessel values were ball-enclosed. An entrywise radius 10^{-10} was added, larger than the analytic quadrature remainder. Interval $LDL^\top$ succeeds after subtracting $10^{-3}I$ in the even sector and $5 \times 10^{-2}I$ in the odd sector. The head–tail couplings are bounded by

$$1.4084884 \times 10^{-9} \quad \text{and} \quad 2.2191643 \times 10^{-10} \quad (5.4)$$

in the even and odd sectors, and the tail deviations below β_* by 9.7777×10^{-24} and 2.4269×10^{-25} . Consequently, by the Weyl inequality applied to the off-diagonal blocks,

$$Q_{0, \log 2}[f] \geq \begin{cases} 9.9999859 \times 10^{-4} \|f\|_2^2, & f \text{ even,} \\ 4.99999998 \times 10^{-2} \|f\|_2^2, & f \text{ odd.} \end{cases} \quad (5.5)$$

The computation is independent of zeta zeros and RH. At $\omega = 0$ it is a certified reproduction of the classical first-window Weil positivity result, not a claim of a new central theorem; see also the screw-form formulation [3] and recent compact-window and Galerkin investigations [4, 5, 12, 13]. The new part here is the parameter-uniform off-center deformation and its use as the density in the shift-energy identity.

There is no conflict between the present first-slab boundary and the larger certified windows in [5]. That paper writes $\text{supp } f \subset [-L_Z, L_Z]$ and its prime sum contains $\log n < 2L_Z$. Our centered interval has total length L , so the conventions are related by $L = 2L_Z$. Thus the classical prime-free cutoff $2L_Z \leq \log 2$ is exactly our $L \leq \log 2$, while the certificate in [5] at $L_Z = 0.8$ is prime-active and concerns the full Weil form, not the archimedean continuation studied here.

6 Entire shift expansion

6.1 Analytic profile

Put $S(t) = tR(t)$. Then

$$S(t) = te^{t/2} - \frac{1}{2}e^{-3t/2} \frac{t}{\sinh t}, \quad (6.1)$$

which is analytic in a disk containing $[0, \log 2]$. Since $\cosh(\omega t) - 1 = \sum_{k \geq 1} \omega^{2k} t^{2k} / (2k)!$, the perturbation is even and entire in ω :

$$C_{\omega, L} = \sum_{k \geq 1} \omega^{2k} T_{k, L}, \quad (T_{k, L})(x, y) = \frac{|x - y|^{2k} R(|x - y|)}{(2k)!}. \quad (6.2)$$

6.2 Exact Legendre correlations

Let m, n have the same parity. For $0 \leq u \leq 1$ define

$$C_{mn}(u) = \frac{1}{2} \int_{-1}^{1-2u} [P_m(r+2u)P_n(r) + P_n(r+2u)P_m(r)] dr, \quad (6.3)$$

a polynomial with rational coefficients. If $S(t) = \sum_{d \geq 0} s_d t^d$, then

$$(T_{k,L})_{mn} = \frac{\sqrt{(2m+1)(2n+1)}}{(2k)!} \sum_j [u^j] C_{mn}(u) \sum_d \frac{s_d L^{d+2k}}{d+2k+j}. \quad (6.4)$$

Thus every finite perturbation matrix is reduced to exact rational Legendre algebra and a scalar analytic series.

6.3 Kernel and Taylor remainder bounds

The profile R changes sign once on $(0, \log 2)$, at

$$t_0 = \log \rho = 0.2811995743229618465 \dots, \quad \rho^3 = \rho + 1, \quad (6.5)$$

where $\rho = 1.3247179572 \dots$ is the plastic number. We locate the actual maximum row sum rather than doubling an endpoint row. Put

$$h(t) = (\cosh(t/2) - 1)R(t), \quad q = e^{t/2}. \quad (6.6)$$

Then

$$h(t) = \frac{(q-1)(q^6 - q^2 - 1)}{2q^2(q+1)(q^2+1)}, \quad (6.7)$$

and dh/dt has the sign of

$$N(q) = q^{10} + q^9 + q^8 + q^7 - q^6 - q^5 + q^4 - 2q^3 - q^2 - q - 1. \quad (6.8)$$

Indeed,

$$N'(q) = (10q^9 - 6q^5) + (9q^8 - 5q^4) + (8q^7 - 6q^2) + (7q^6 - 2q) + (4q^3 - 1) > 0 \quad (6.9)$$

for $q \geq 1$, since each bracket is nonnegative at $q = 1$ and increasing. Also $N(1) = -2 < 0$ and $N(\sqrt{\rho}) = 3\rho^2 - 1 > 0$. Thus h has a unique minimum t_m before t_0 and is increasing after t_m . The Arb check encloses

$$\begin{aligned} t_m &= 0.1416454811943886 \dots, \\ h(t_m) &= -0.0044442714714077 \dots, \\ h(\log 2 - t_0) &= 0.0126146638556245 \dots, \end{aligned} \quad (6.10)$$

with $h(\log 2 - t_0) + h(t_m) > 0.00817039$.

Lemma 6.1 (endpoint row maximum). *For $0 < L \leq \log 2$ and $0 \leq \omega \leq \frac{1}{2}$,*

$$\|C_{\omega,L}\| \leq \int_0^{\log 2} |h(t)| dt \leq 0.012293050401142320 < 0.012294. \quad (6.11)$$

Proof. Take first $L = \log 2$ and $\omega = \frac{1}{2}$, so that $|c_\omega(x, y)| = |h(|x - y|)|$. For a row indexed by $x \in I_L$, let its left and right available lengths be a, b , with $a + b = L$, and assume $a \geq b$. Its absolute row integral is $F(a) + F(b)$, where $F(r) = \int_0^r |h(t)| dt$, and $\frac{d}{db}[F(L - b) + F(b)] = |h(b)| - |h(a)|$. Note that $a \geq L/2 = 0.3466\dots > t_0 = 0.2812\dots$, so $a \leq t_0$ cannot occur and $h(a) > 0$ always. If $b \geq t_0$, monotonicity of h on $[t_m, \infty)$ gives $h(a) \geq h(b) \geq 0$. If $b \leq t_0$, then $h(b) \geq h(t_m)$ and $h(a) \geq h(\log 2 - t_0)$ since $a \geq \log 2 - t_0 > t_m$, so the displayed interval check gives $h(a) + h(b) > 0$, again implying $|h(a)| \geq |h(b)|$. Hence the absolute row integral is nonincreasing in b on $[0, L/2]$ and is maximal at an endpoint row, where it equals $\int_0^{\log 2} |h|$. The Schur test for the symmetric kernel gives the first inequality; sign-split Arb integration gives the second. The bound is uniform for $0 \leq \omega \leq \frac{1}{2}$ and $0 < L \leq \log 2$, since $|c_\omega|$ is pointwise increasing in ω and restriction to a shorter centered interval can only shorten rows. $\square$

Truncating the shift expansion after $\omega^8 T_4$ gives

$$\sup_{0 \leq \omega \leq 1/2} \left\| C_{\omega, L} - \sum_{k=1}^4 \omega^{2k} T_{k, L} \right\| < 1.54 \times 10^{-12}. \quad (6.12)$$

The primary Arb enclosure, obtained from a conservative absolute-profile Cauchy bound rather than a sharp sign split, is $1.532967865231399 \times 10^{-12}$.

7 Computer-assisted generator positivity

7.1 Parameter boxes

The interval $0 \leq \omega \leq \frac{1}{2}$ is divided into 256 equal boxes. For each box and parity sector the calculation forms

$$Q_0 + [\omega^2]T_1 + [\omega^4]T_2 + [\omega^6]T_3 + [\omega^8]T_4, \quad (7.1)$$

where every occurrence of ω is represented by the same interval parameter. This retains correlation between all matrix entries.

The sixteen-mode head is combined with the infinite Legendre tail. The central tail lower bound is reduced by the full perturbation norm:

$$d_{\text{tail}} \geq \beta_* - \delta_{\text{central}} - 0.012294 > 1.5176929819194764. \quad (7.2)$$

Here δ_{central} is not set to zero: its certified values are 9.7777×10^{-24} in the even sector and 2.4269×10^{-25} in the odd sector, and therefore disappear at the displayed decimal precision. The head–tail coupling is bounded by

$$\kappa = \kappa_{\text{central}} + 0.012294, \quad (7.3)$$

with κ_{central} from (5.4). The resulting worst Schur loss κ^2/d_{tail} is $9.95869865858903 \times 10^{-5}$ (even) and $9.95869673623490 \times 10^{-5}$ (odd).

7.2 Primary Arb result

For every box, interval LDL^T proves positivity of the shifted head after subtracting the Schur loss and Taylor remainder. All 512 parity-box problems pass. The minimum certified quantities are as follows.

Quantity	Even	Odd
Weyl–Schur margin m	$6.116350109501 \times 10^{-4}$	$5.261670009275 \times 10^{-2}$
tail lower bound d_{tail}	1.51769298191948	1.51769298191948
Schur loss κ^2/d_{tail}	$9.95869865858903 \times 10^{-5}$	$9.95869673623490 \times 10^{-5}$

Across both sectors the smallest interval $LDL^\top$ pivot is $2.326811914886 \times 10^{-2}$. The margin m is, by construction, a certified lower bound for $\lambda_{\min}(\text{head}) - \kappa^2/d_{\text{tail}}$ on each box, with the interval widening of the box, the entrywise radii, and the Taylor remainder (6.12) already subtracted.

7.3 From the certified margin to coercivity on the form domain

The certificate is a statement about a 16×16 head, an infinite tail, and their coupling. The following elementary lemma converts it into a coercivity bound for the full form on $\mathcal{D}_{\log}$, which is the form in which Section 8.6 consumes it.

Lemma 7.1 (block-Schur coercivity). *Let Q be a closed lower-bounded form with domain $\mathcal{D} \subset \mathcal{H}$, let $\mathcal{H} = \mathcal{H}_h \oplus \mathcal{H}_t$ with $\mathcal{H}_h$ finite-dimensional and $\mathcal{H}_h \subset \mathcal{D}$, and suppose that for $f_h \in \mathcal{H}_h$ and $f_t \in \mathcal{D} \cap \mathcal{H}_t$,*

$$Q[f_h] \geq \lambda \|f_h\|^2, \quad Q[f_t] \geq d \|f_t\|^2, \quad |Q[f_h, f_t]| \leq \kappa \|f_h\| \|f_t\|,$$

with $d > 0$. Then $Q[f] \geq c \|f\|^2$ for all $f \in \mathcal{D}$ and every $0 \leq c < \min(\lambda, d)$ with $(\lambda - c)(d - c) \geq \kappa^2$. In particular, if $\lambda \geq m + \kappa^2/d$ with $0 < m < d$, then $Q \succeq cI$ for

$$c = m - \frac{\kappa^2 m}{d(d - m)}. \quad (7.4)$$

Proof. For $f = f_h + f_t$ with $f_h \in \mathcal{H}_h$ and $f_t = f - f_h \in \mathcal{D} \cap \mathcal{H}_t$,

$$Q[f] = Q[f_h] + 2 \operatorname{Re} Q[f_h, f_t] + Q[f_t] \geq \lambda \|f_h\|^2 - 2\kappa \|f_h\| \|f_t\| + d \|f_t\|^2,$$

and the quadratic form on the right dominates $c(\|f_h\|^2 + \|f_t\|^2)$ exactly when the matrix $\begin{pmatrix} \lambda - c & -\kappa \\ -\kappa & d - c \end{pmatrix}$ is positive semidefinite, i.e. when $\lambda \geq c$, $d \geq c$, and $(\lambda - c)(d - c) \geq \kappa^2$. For the second statement put $c = m - \delta$ with $\delta = \kappa^2 m / (d(d - m))$; then $(\lambda - c)(d - c) \geq (\kappa^2/d + \delta)(d - m + \delta) \geq \kappa^2 + \delta(d - m) - \kappa^2 m/d = \kappa^2$. $\square$

Remark 7.2. The naive reading “ $Q \succeq mI$ ” is *not* implied by the certificate: the Schur complement of the tail at level c is $\kappa^2/(d - c)$, not κ^2/d , and the difference costs $\kappa^2 m / (d(d - m))$. Because $d_{\text{tail}} \approx 1.5177 \gg m$ the correction is small— 4.0×10^{-8} in the even sector and 3.6×10^{-6} in the odd sector—but it is the reason Theorem B states $c_* = 6.1159 \times 10^{-4}$ rather than the certified margin itself. With the data of Section 7.2, (7.4) gives $c_{\text{even}} = 6.11594 \dots \times 10^{-4}$ and $c_{\text{odd}} = 5.26131 \dots \times 10^{-2}$.

Proof of Theorem B. Fix a parity sector and $L = \log 2$. Take $\mathcal{H}_h$ to be the span of the sixteen head Legendre modes of that parity, $\mathcal{H}_t$ its orthogonal complement in the sector, and $\mathcal{D} = \mathcal{D}_{\log}(I_{\log 2})$ intersected with the sector. For each box, the certificate gives $\lambda \geq m + \kappa^2/d_{\text{tail}}$ on the head, d_{tail} on the tail by (7.2), and κ by (7.3). Lemma 7.1 yields $Q_{\omega, \log 2} \succeq c_{\text{even}}I$ and $Q_{\omega, \log 2} \succeq c_{\text{odd}}I$ on the two sectors, for every ω in the box; the union of the boxes is $[0, \frac{1}{2}]$ and the two sectors are orthogonal and Q -invariant, so (1.8) holds at $L = \log 2$ with $c_* = \min(c_{\text{even}}, c_{\text{odd}}) \geq 6.1159 \times 10^{-4}$.

If $0 < L' < \log 2$, zero extension identifies $L^2(I_{L'})$ with a closed subspace of $L^2(I_{\log 2})$ and preserves parity. The representations (4.7)–(4.8) depend on f only through $\hat{f}$ and

the functionals $\mathcal{C}_c(f), \mathcal{S}_c(f)$, none of which depends on the horizon, so $Q_{\omega, L'}$ is exactly the restriction of $Q_{\omega, \log 2}$ to that subspace, and $\mathcal{D}_{\log}(I_{L'})$ is the corresponding restriction of the form domain. Positivity with the same constant therefore passes to every shorter centered interval. In the causal coordinates of Section 2, the same statement says that causality makes $V_{\omega, L'}$ the finite-horizon compression of $V_{\omega, \log 2}$. $\square$

7.4 Endpoint safety

The certificate includes $\omega = \frac{1}{2}$. No limiting digamma evaluation is used there. The bounded-kernel representation $Q_{\omega} = Q_0 + C_{\omega}$ continues analytically through the endpoint and carries the boundary delta contribution that would be lost by a pointwise multiplier limit. Explicitly, that contribution is $-\pi\delta_0$ in the multiplier, or $-\frac{1}{2}|f|^2$ in the quadratic form, as stated in Lemma 4.1.

7.5 Exact role of the computer

The following inputs are analytic:

1. the transfer identity (4.16) for $a_{\omega} - a_0$;
2. the bounded-kernel formula (4.11);
3. the common logarithmic form domain (4.17);
4. the rational Legendre correlation formula (6.4);
5. the scalar Cauchy remainder for S ;
6. the kernel Schur bound of Lemma 6.1;
7. the spherical-Bessel infinite-tail estimates;
8. the block-Schur reduction of Lemma 7.1.

The computer performs interval evaluation of the finite matrices and interval LDL^T on the 512 boxes. The theorem does not infer positivity from a floating grid.

7.6 Independent implementation audit

A clean-room audit implementation imports neither the primary generator code nor the central-certificate code. Its independent paths are:

- arbitrary-precision digamma and Bessel evaluation rather than Arb;
- separately generated 16- and 48-point quadrature rules;
- the closed binomial formula for Legendre polynomials rather than recurrence;
- the Bernoulli expansion of $t/\sinh t$ rather than reciprocal-series inversion;
- direct nonlinear-kernel integrals at selected entries;
- a custom 80-digit directed-rounding `Decimal` interval class; and
- an independent interval LDL^T implementation.

The maximum central quadrature discrepancy is 1.6931×10^{-15} . The maximum difference between the independently reconstructed and primary floating minimum eigenvalues is 4.3755×10^{-13} . The maximum discrepancy between direct nonlinear-kernel integration and the degree-four Taylor reconstruction is 1.5929×10^{-13} . All 512 boxes pass. The independently recovered minimum margins are $6.116514636999 \times 10^{-4}$ and $5.261670255041 \times 10^{-2}$, and the minimum directed-interval pivot is $2.326829938890 \times 10^{-2}$.

The audit assigns every central entry a radius 5×10^{-10} , over five orders of magnitude larger than the observed independent quadrature discrepancy. Its final margins are nevertheless slightly larger than the primary margins because the two programs propagate the shift-polynomial coefficient uncertainty differently. On the weakest even box, the audit mid-point is 1.57×10^{-9} lower but its row-sum radius is 1.80×10^{-8} smaller, producing a net margin increase of 1.65×10^{-8} . On the weakest odd box, the corresponding changes are -2.12×10^{-8}

in the midpoint and -2.36×10^{-8} in the radius, producing a net increase of 2.46×10^{-9} . Thus the audit's larger fixed central radius is more than offset by its tighter directed interval propagation of the parameter-dependent coefficients; the two final intervals are independent enclosures and are not asserted to be nested. This is a strong independent reproduction. Its one qualification is that the second special-function path is arbitrary-precision rather than a second formally validated interval special-function library. The original Arb calculation remains the formal special-function enclosure. The endpoint-row-maximum *argument* of Lemma 6.1 is inherited by the audit from the primary implementation; only the scalar integral in (6.11) was recomputed independently.

8 The shift-energy identity

8.1 Weighted Hardy realization

Fix $\eta > 1$ and consider $\mathcal{H}_\eta = L^2(\mathbb{R}_+, e^{-2\eta t} dt)$. The Laplace transform on $\text{Re } p = \eta$ identifies this space with a Hardy boundary space. Multiplication by K_s defines a bounded causal operator T_s , and

$$V_{s,L} = P_L T_s E_L, \quad (8.1)$$

where E_L is zero extension and P_L is restriction. The choice $\eta > 1$ keeps the line $\text{Re } p = \eta$ to the right of the pole $p = \alpha < \frac{1}{2}$ of (3.5).

Uniform Stirling estimates on the fixed right half-plane give

$$K_s(\eta + i\tau) = O((2 + |\tau|)^{-s}) \quad (8.2)$$

uniformly for $0 \leq s \leq \frac{1}{2}$. For every $\varepsilon > 0$ and $m = 0, 1, 2$,

$$\partial_s^m K_s(\eta + i\tau) = O((1 + \log(2 + |\tau|))^m (2 + |\tau|)^{-\varepsilon}) \quad (8.3)$$

uniformly for $\varepsilon \leq s \leq \frac{1}{2}$. Also $a_s(\eta + i\tau) = O(1 + \log(2 + |\tau|))$, so $a_s K_s$ is a bounded multiplier for every positive s . Consequently $s \mapsto V_{s,L}$ is continuously differentiable in operator norm on every compact positive-shift interval, and

$$V'_{s,L} = -A_{s,L} V_{s,L}. \quad (8.4)$$

Although $A_{s,L}$ is unbounded, the product $A_{s,L} V_{s,L}$ is bounded. The weighted Hardy realization is used only to prove this adjoint-free derivative identity. On the finite interval the weighted and unweighted L^2 norms are equivalent, so the identity and its norm-differentiability transfer to operators on unweighted $L^2(0, L)$. Every adjoint from Section 8.4 onward is taken with respect to that unweighted inner product.

8.2 Strong limit at zero

For each fixed p on $\text{Re } p = \eta$, $K_s(p) \rightarrow 1$ as $s \downarrow 0$. The uniform multiplier bound and dominated convergence give

$$V_{s,L} f \longrightarrow f \quad \text{in } L^2(0, L) \quad (8.5)$$

for every $f \in L^2(0, L)$. This convergence is strong, not in operator norm: the high-frequency decay $K_s(\eta + i\tau) = O(|\tau|^{-s})$ prevents uniform multiplier convergence to one.

8.3 Smoothing into the common form domain

For $0 < r < s$, $\sup_\tau (1 + |\tau|)^r |K_s(\eta + i\tau)| < \infty$. Choose $r < \min(s, \frac{1}{2})$. The full causal multiplier maps L^2 into H^r , and restriction followed by zero extension is bounded on H^r for $r < \frac{1}{2}$. Since $\log(2 + |\tau|) \leq C_r(1 + |\tau|)^{2r}$, we obtain

$$V_{s,L}L^2(0, L) \subset \mathcal{D}_{\log}(I_L). \quad (8.6)$$

By closure from the smooth core,

$$Q_{s,L}[V_{s,L}f] = \operatorname{Re}\langle A_{s,L}V_{s,L}f, V_{s,L}f \rangle. \quad (8.7)$$

8.4 Energy identity away from zero

Let $D_{s,L} = I - V_{s,L}^*V_{s,L}$. For $0 < \varepsilon < \omega \leq \frac{1}{2}$, operator-norm differentiation gives

$$\frac{d}{ds}\langle D_{s,L}f, g \rangle = -\langle V_{s,L}'f, V_{s,L}g \rangle - \langle V_{s,L}f, V_{s,L}'g \rangle = 2Q_{s,L}[V_{s,L}f, V_{s,L}g]. \quad (8.8)$$

The right-hand side represents a norm-continuous bounded operator on $[\varepsilon, \omega]$. Therefore

$$D_{\omega,L} - D_{\varepsilon,L} = 2 \int_{\varepsilon}^{\omega} V_{s,L}^* Q_{s,L} V_{s,L} ds, \quad (8.9)$$

with an operator-norm integral.

8.5 Passage to zero

By the strong approximate-identity result (8.5), $D_{\varepsilon,L} \rightarrow 0$ strongly. Theorem B makes the integrand nonnegative. Letting $\varepsilon \downarrow 0$ gives

$$D_{\omega,L} = 2 \int_0^{\omega} V_{s,L}^* Q_{s,L} V_{s,L} ds \succeq 0, \quad (8.10)$$

where the improper integral is the monotone strong limit of the norm integrals over $[\varepsilon, \omega]$. Equivalently,

$$\langle D_{\omega,L}f, f \rangle = 2 \int_0^{\omega} Q_{s,L}[V_{s,L}f] ds. \quad (8.11)$$

8.6 Strictness

Suppose $\langle D_{\omega,L}f, f \rangle = 0$. Theorem B is uniform in the shift and gives, for the maximal horizon and hence for every compression,

$$Q_{s,L}[g] \geq c_* \|g\|_2^2, \quad c_* = 6.1159 \times 10^{-4}, \quad 0 \leq s \leq \frac{1}{2}, \quad g \in \mathcal{D}_{\log}(I_L). \quad (8.12)$$

This is a bound on the whole form domain, not only on the certified head: Lemma 7.1 converts the certified head margin into it. The scalar integrand in (8.11) is measurable and nonnegative. Its integral can vanish only if $Q_{s,L}[V_{s,L}f] = 0$ for almost every $s \in (0, \omega)$. The uniform bound implies $V_{s,L}f = 0$ for almost every such s . Choose a sequence from this full-measure set with $s_n \downarrow 0$. The strong limit $V_{s_n,L}f \rightarrow f$ then yields $f = 0$. Hence $D_{\omega,L} > 0$.

For fixed $s > 0$, the weak singularity $\kappa_s(t) = O(t^{s-1})$ is integrable. Truncating κ_s away from $t = 0$ gives Hilbert–Schmidt Volterra operators, while Young’s inequality shows convergence in operator norm. Thus $V_{s,L}$ is compact. If $\|V_{\omega,L}\| = 1$, the compact positive operator $V_{\omega,L}^*V_{\omega,L}$ has eigenvalue one, contradicting the positive definiteness of $D_{\omega,L}$. Therefore $\|V_{\omega,L}\| < 1$. This proves Theorem C and, with (2.10), Theorem A on $0 < \omega \leq \frac{1}{2}$.

9 Consequences and interpretations

9.1 Reflection positivity

Since $H_{\omega,L} = V_{\omega,L}R_L$ is self-adjoint and $\|H_{\omega,L}\| = \|V_{\omega,L}\| < 1$, we have $I \pm H_{\omega,L} \succ 0$. This is reflection positivity for the two finite-volume Hankel channels, in the operator-positivity sense inherited from the Osterwalder–Schrader framework [9]. Shorter horizons follow either from the same proof or by compression from $L = \log 2$.

9.2 Shift-radial Gram factor

Let $\mathfrak{K}_{\omega,L} = L^2((0, \omega); L^2(0, L))$ and define $(\mathcal{G}_{\omega,L}f)(s) = \sqrt{2}Q_{s,L}^{1/2}V_{s,L}f$. The energy identity gives

$$2 \int_0^\omega \|Q_{s,L}^{1/2}V_{s,L}f\|^2 ds = \langle D_{\omega,L}f, f \rangle \leq \|f\|^2. \quad (9.1)$$

Hence $\mathcal{G}_{\omega,L}$ is bounded and $D_{\omega,L} = \mathcal{G}_{\omega,L}^* \mathcal{G}_{\omega,L}$. The boundary defect is therefore the accumulated positive energy along the shift coordinate.

9.3 Context: the classical shifted range $\omega > \frac{1}{2}$

For $\omega > \frac{1}{2}$ the function Θ_ω is inner unconditionally: $\zeta(s) \neq 0$ for $\operatorname{Re} s \geq 1$, so [1, Prop. 1.2] applies with $\omega_0 = \frac{1}{2}$. Global innerness makes H_ω an isometry [1, Lemma 4.1] and gives the non-strict estimate $\|V_{\omega,L}\| \leq 1$ for every finite section. Suzuki goes further: for $\omega > \frac{1}{2}$ and every $a > 1$ he proves [1, Lemma 4.4] that $\|H_{\omega,a}\| < 1$, so that $1 \pm H_{\omega,a}$ are invertible. The strictness there comes from a support lemma [1, Lemma 4.3] of Paley–Wiener type—for $\omega > \frac{1}{2}$, $H_\omega P_a f$ cannot have compact support unless it vanishes—combined with the isometry and the compactness of $H_{\omega,a}$. In the normalization of Section 2.2 this covers all $\omega > \frac{1}{2}$ and all $L = 2 \log a > 0$, at every depth, prime layers included.

Theorem A is the complementary statement on the critical interval. It covers $0 < \omega \leq \frac{1}{2}$, where no zero-free input is available and Suzuki’s support lemma is not stated, at the prime-free depths $L \leq \log 2$. At the endpoint $\omega = \frac{1}{2}$, where [1, Prop. 1.2] gives nothing and [1, Lemma 4.4] is not stated, Theorem A supplies strictness directly, without invoking innerness. We make no additional finite-section claim for $\omega > \frac{1}{2}$ beyond citing [1], and the shift-radial Gram factor of Section 9.2 is the only factorization theorem asserted in this paper. Extending the certificate of Theorem B to $\omega \in [\frac{1}{2}, \omega_1]$ would give, via (1.9), $D_{\omega,L} \succeq D_{1/2,L} \succ 0$ and hence an independent Paley–Wiener-free proof of strict contractivity on the first slab in that range; floating diagnostics (Section 12.3) suggest this is available up to $\omega_1 \approx 0.86$ in the even sector, but no such extension is claimed.

10 Relation to the broader shifted-zeta program

10.1 String and screw-function realizations

The shifted string Weyl function is

$$q_\omega(z) = \frac{1}{\sqrt{-z}} \frac{\xi'}{\xi} \left(\frac{1}{2} + \omega + \sqrt{-z} \right), \quad \operatorname{Re} \sqrt{-z} > 0. \quad (10.1)$$

Under the corresponding shifted zero-free hypothesis this is a Stieltjes function and defines a positive Kreĭn string. Its spectral density is

$$\rho_\omega(u) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi} \left(\frac{1}{2} + \omega + iu \right). \quad (10.2)$$

The associated Källén–Lehmann superposition is reflection positive. This is an exact inverse-spectral reformulation of the shifted zero-free condition (compare the shifted screw functions Ψ_ω of [2, Sect. 11]), but it is downstream of positivity. By contrast, the first-slab Volterra system is specified directly from gamma and prime coefficients and is positive without a zero-free assumption.

If $A_\omega(t)$ denotes Suzuki’s integrated radial profile and $\mathcal{E}_\omega = \int_0^\infty |1 - A_\omega(t)|^2 dt$, then on the inner side

$$\mathcal{E}_\omega = 2 \frac{\xi'}{\xi} \left(\frac{1}{2} + \omega \right), \quad \rho_\omega(0) = \frac{\mathcal{E}_\omega}{2\pi}, \quad \lim_{x \rightarrow \infty} \frac{m_\omega^{\text{str}}(x)}{x} = \frac{4}{\mathcal{E}_\omega^2}. \quad (10.3)$$

Thus the screw margin, phase density, boundary energy, and infrared string density are different transforms of one scalar quantity. These companion statements are developed in [14, 15, 16] and are not used here.

10.2 Modular endpoint

Let $\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ and $\varphi(s) = \Lambda(2s - 1)/\Lambda(2s)$. At $\omega = \frac{1}{2}$,

$$\Theta_{1/2}(u) = \frac{u - i}{u + i} \varphi\left(\frac{1 - iu}{2}\right). \quad (10.4)$$

The critical shift endpoint is therefore the modular cusp scattering coefficient completed by one elementary Blaschke channel, in the standard scattering-theoretic sense [10]. This supplies a geometric anchor, but physical-line unitarity is unconditional; the RH-sensitive content remains off-line causal analyticity and contractivity.

10.3 Radial reflection positivity and holographic language

The term *radial reflection positivity* is literal here: one logarithmic coordinate is reflected, and the boundary form is the contraction defect of a causal radial propagator. The identity $D_{\omega,L} = 2 \int_0^\omega V_{s,L}^* Q_{s,L} V_{s,L} ds$ is a positive-bulk reconstruction of the boundary defect. The word *holographic* is useful only in this transfer-system sense. The construction is not presently a local Poincaré–Einstein geometry, and the boundary object is not generally a homogeneous CFT two-point function. A closer PDE precedent is Burnol’s causal Klein–Gordon realization of the classical Hankel transform [6].

10.4 Where arithmetic first enters

For $\log 2 < L < \log 3$ the kernel becomes

$$k_\omega(t) = \kappa_\omega(t) + \frac{c_\omega(2)}{\sqrt{2}} \kappa_\omega(t - \log 2). \quad (10.5)$$

The new term is a delayed copy of the archimedean impulse. Extending the positive energy identity across this update is the first genuinely arithmetic problem. The present theorem supplies the base storage operator against which that delayed layer must be tested.

11 Alternative checks and superseded proof routes

11.1 Compact Mosco limit

The normalized defect channels $\omega^{-1} \langle f, (I \mp H_{\omega,L}) f \rangle$ converge as $\omega \downarrow 0$ to the even and odd restrictions of $Q_{0,L}$. Uniform logarithmic coercivity prevents high-frequency escape,

and compact Mosco convergence combines with the central strict gaps to give a qualitative small-shift theorem. This route remains an independent analytic explanation of the singular endpoint. It is no longer needed for the complete interval theorem, because the shift-energy proof uses the strong multiplier limit $V_s \rightarrow I$ directly.

11.2 Direct transfer certificates

Earlier calculations certified the defect channels directly through a parity-scaled matrix-valued Taylor model and centered endpoint charts. These computations supplied substantial numerical and interval evidence, but the finite- ω defect matrices mix parity and require more elaborate conditioning than the generator form. The generator certificate is preferable because the generator remains parity diagonal; all shift dependence is a compact kernel entire in ω^2 ; the degree-four remainder is below 1.54×10^{-12} ; the full interval, including both endpoints, is covered uniformly; and the energy identity converts generator positivity into the desired defect inequality. The earlier certificates are retained in the research archive as independent cross-checks, but not as part of the main proof.

12 Limitations and next problems

12.1 A fully dual-library certificate

The clean-room audit independently reconstructs the mathematics and interval LDL^T calculation. Its central special-function path uses arbitrary precision with a conservative audit radius rather than a second formal interval library. A publication repository would be stronger if the central digamma/Bessel quadrature were also implemented in a second validated interval environment. The current agreement makes this an audit enhancement, not a practical blocker.

12.2 The second slab

The immediate mathematical target is the delayed $n = 2$ layer. Natural approaches include differentiating the complete delayed propagator, deriving a storage update or Redheffer-star-product inequality, and testing whether the delayed layer preserves passivity.

The archimedean continuation cannot be assumed positive throughout this next slab. A 24-mode floating diagnostic, independently rerun but not promoted to an interval certificate, places the first odd and even crossings of the prime-free generator near

$$L_{\text{odd}}^* \approx 0.7429, \quad L_{\text{even}}^* \approx 0.8199. \quad (12.1)$$

In particular, the odd sector is already negative at $L = 0.75$, only about seven percent beyond $\log 2$, and both sectors are negative before $\log 3$. These values concern the prime-free archimedean continuation; they do not contradict the prime-active full-Weil certificates of [5]. They change the logic of the second-slab project: the delayed $n = 2$ term cannot be treated over the whole slab merely as a small correction to a positive base. It must be analyzed as a possible source of the missing positivity, or as the source of a precise obstruction.

At the fixed first-slab horizon $L = \log 2$, the same floating diagnostic places the first even generator crossing near $\omega = 0.8611$ and the odd crossing near $\omega = 1.8047$. Those estimates suggest a possible extension of the generator certificate beyond the critical interval (Section 9.3), but such an extension is outside Theorem B and should be stated only after a separate interval run.

The next steps are therefore: derive the exact differentiated $n = 2$ delayed generator; isolate its signed contribution relative to the archimedean continuation; seek a storage or

Redheffer inequality that allows the delayed layer to restore, rather than merely preserve, positivity; certify representative horizon slices in interval arithmetic; and identify the first rigorous obstruction if the repair mechanism fails.

12.3 Euler-product network and geometry

The coefficient train satisfies

$$\sum_{n \geq 1} \frac{c_\omega(n)}{\sqrt{n}} n^{-s} = \frac{\zeta(s + \frac{1}{2} - \omega)}{\zeta(s + \frac{1}{2} + \omega)}. \quad (12.2)$$

A passive-network composition law reflecting this Euler product would be a natural global continuation of the first-slab theorem. Separately, one may seek a local PDE or boundary-triple realization of the completed-gamma quotient whose energy estimate produces $Q_{s,L}$ directly.

13 Verification status

This section replaces the private author-verification notice of the working drafts. It records what has been checked, by what means, and what has not. The manuscript has been through two adversarial referee-style review rounds with independent numerical recomputation of every reported constant, and one clean-room reimplementing of the certificate (Section 7.6). No error survives in the main proof chain. What the manuscript has *not* had is line-by-line derivation of the central analytic steps by a human reader who will stand behind them; the deposit of this preprint is not a claim that this has happened.

Independently reproduced from the printed definitions

Two further recomputations, independent of both the primary Arb code and the clean-room audit—one made during the referee-style review rounds, one made while preparing this preprint—reproduced the following from the definitions printed here: all algebraic identities in Sections 2.2, 3.1, 3.3, 4.1, 6.1, 6.2, 10.2 and 10.4; the Laplace transform of the explicit impulse (3.6) against (3.3) (to 1.5×10^{-17} at $\omega = \frac{1}{2}$ and 8×10^{-11} at $\omega = 0.3$, the latter limited by quadrature at the $t^{\omega-1}$ singularity); the multiplier identity of Proposition 4.2 to 30 digits at $\omega = 0.1, 0.3$ and $\frac{1}{2}$; each step of its proof (digamma integral representation, cancellation of the divergent summands, the rational kernel $4 \cosh(t/2)(\cosh \omega t - 1)$, and the bridge identity for R); the endpoint Lemma 4.1, whose corrected formula agrees with $Q_0 + C_{1/2}$ for test functions to the quadrature floor ($\sim 10^{-8}$) while the uncorrected pointwise-limit formula is off by 0.2402; the complete chain of Section 6.3 (t_0 , the closed form of h , the sign analysis of N , t_m , $h(t_m)$, $h(\log 2 - t_0)$, $\int_0^{\log 2} |h| = 0.012293050401142319559\dots$, and the inequality $L/2 > t_0$); the tail bound (7.2) and both Schur losses to 17 digits; the Taylor remainder $1.532967865231399 \times 10^{-12}$; the head minima and margins of Section 7.2 by an independent sixteen-mode Legendre reconstruction, consistent with the reported values to within the expected interval-widening penalty (2.26×10^{-5} even, 8.65×10^{-6} odd); the coercivity constants of Remark 7.2; and all reference metadata.

Checked against Suzuki’s statements

The transfer normalization (3.3) and the layer decomposition (2.5) were checked against the displayed statements of [1] as follows: Suzuki’s factor $\gamma(s)$ is $\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2) = \Gamma_\infty(s)$; his Mellin formula [1, Sect. 3.1] for g_ω is (3.3) after $s = p + \frac{1}{2}$; (2.5) is [1, eq. (2.3)] in

logarithmic coordinates; and [1, Prop. 2.1] is consistent with these two facts and the Dirichlet series of c_ω , with matching domains of convergence. The identification of $H_{\omega,a}$ with the centered interval of length $2 \log a$ (Section 2.2) is derived from [1, eqs. (2.5)–(2.6)]. These comparisons were made against automated readings of the published text and should be confirmed by a human reader against the paper itself; that confirmation is the single most important outstanding check, since a factor or shift there would move the $\log 2$ threshold.

Not yet human-verified

- (a) *Elementary derivations on paper*: Proposition 4.2 including its endpoint extension; Lemma 4.1; Lemma 6.1; the Volterra identities (2.10).
- (b) *Functional analysis of Section 8*: the operator-norm C^1 dependence of $s \mapsto V_{s,L}$ (the multiplier bounds of Section 8.1 are stated with $O(\cdot)$ and no constants, and are the weakest-supported claims in the manuscript); the weighted-to-unweighted transfer; the closure step (8.7); the improper strong-operator integral of Section 8.5. Failure of these would leave Theorem C unproved and Theorem A resting on the routes of Section 11, not make Theorem A false.
- (c) *Analytic number theory*: human confirmation of the normalization chain against [1] (above); an originality search for the off-center kernel identity of Proposition 4.2 and the shift-radial energy identity against [4, 5, 12, 13] and newer work in this active neighborhood.
- (d) *Repository*: the inherited central certificate (the Binet bound $w_0(\tau) \geq \beta_*$ for $|\tau| \geq 30$ and the spherical-Bessel Legendre-tail majorants behind (5.4)), which both the primary and audit implementations take as given; the interval $LDL^\top$ and box-construction code; a full primary-versus-audit row diff rather than a comparison of summary minima; and a version-pinned rerun of both implementations in clean environments with archived logs.

Optional strengthenings (not verification): promote the positivity-horizon crossings of Section 12.3 to interval certificates; extend the box sweep to $\omega \approx 0.85$ (Section 9.3); double the box count to 512, since the box radius (2.42×10^{-5}) is now the second-largest erosion of the even margin after the Schur loss.

A Normalization ledger

The manuscript uses

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \Gamma_\infty(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad (\text{A.1})$$

$$\Theta_\omega(z) = \frac{\xi(\frac{1}{2} - \omega - iz)}{\xi(\frac{1}{2} + \omega - iz)}, \quad K_\omega(p) = \frac{\Gamma_\infty(\frac{1}{2} + p - \omega)}{\Gamma_\infty(\frac{1}{2} + p + \omega)}, \quad (\text{A.2})$$

$$Q_{\omega,L} = \frac{1}{2}(A_{\omega,L} + A_{\omega,L}^*), \quad L_{\max} = \log 2, \quad T_0 = \frac{1}{2} \log 2. \quad (\text{A.3})$$

Suzuki's multiplicative compression $H_{\omega,a} = P_a H_\omega P_a$ corresponds to $L = 2 \log a$; the compact-window convention of [5] corresponds to $L = 2L_Z$. For the shifted string,

$$\rho_\omega(u) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\tfrac{1}{2} + \omega + iu), \quad q_\omega(z) = \frac{1}{\sqrt{-z}} \frac{\xi'}{\xi}(\tfrac{1}{2} + \omega + \sqrt{-z}). \quad (\text{A.4})$$

The factor $Q = B/2$, where $B = A + A^*$, is essential when comparing the central Weil certificate, perturbation matrices, and shift-energy identity.

B Reproducibility

The certificate scripts described below are not part of this deposit; they are available from the author on request, and a code release with pinned environments and archived logs is planned once the repository items of Section 13(d) are complete.

B.1 Primary certificate

The primary generator certificate is produced by the script

```
investigation12_offcenter_weil_generator.py,
```

with the inherited central Weil implementation in

```
project_support/investigation7_first_slab_certificate.py:
```

```
PYTHONPATH=project_support:. \
python3 investigation12_offcenter_weil_generator.py \
  --certify --boxes 256 --output INVESTIGATION_12_results.csv
python3 verify_investigation12_results.py
```

Expected summary:

```
certified boxes: 512
kernel norm < 0.012294
Taylor remainder < 1.54e-12
minimum interval LDL pivot > 0.0232681
minimum certified Weyl margin > 0.00061163
```

The 724 result rows consist of 6 Laplace-transform identity checks, 1 kernel norm row, 1 Taylor-remainder row, 202 floating generator samples (101 shifts in two parity sectors), 2 relative-endpoint checks, and 512 certified parameter boxes.

B.2 Independent audit

The audit is produced by the script

```
investigation14_independent_generator_audit.py:
```

```
python3 investigation14_independent_generator_audit.py \
  --primary INVESTIGATION_12_results.csv \
  --output INVESTIGATION_14_results.csv \
  --matrices INVESTIGATION_14_matrices.csv
python3 verify_investigation14_results.py
```

Expected summary:

```
audit rows: 783
matrix rows: 1360
maximum central quadrature discrepancy: 1.693e-15
maximum primary floating discrepancy: 4.375e-13
maximum direct-kernel discrepancy: 1.593e-13
minimum directed-interval margin: 6.116514636999e-04
minimum directed-interval pivot: 2.326829938890e-02
```

The 783 audit rows split as follows: 2 central-quadrature convergence rows, 1 kernel norm, 1 Taylor remainder, 24 Legendre-correlation comparisons, 5 profile-series comparisons, 36 direct-kernel comparisons, 202 primary floating comparisons, and 512 directed-interval boxes. The 1360 matrix rows are the lower triangles of ten 16×16 coefficient matrices: five Taylor matrices in each of two parity sectors, with 136 stored entries per matrix.

B.3 Analytic diagnostics

The shift-energy transfer diagnostics are generated by

`investigation13_operator_diagnostics.py`

and checked by `verify_investigation13_results.py`. They verify the transfer derivative, pointwise multiplier limit, high-frequency regularization, and local kernel tail on a high-precision sample grid. These checks are not ingredients of the analytic proof.

C Proof-status ledger

Statement	Status
First-slab Volterra reduction	exact
Completed-gamma transfer	exact; matches [1, Sect. 3.1]
Sonine convolution identity	exact
Formula (4.2) for a_ω	exact
Fourier-plus-residue endpoint term	exact distributional correction
Bounded-kernel decomposition $Q_\omega = Q_0 + C_\omega$	exact
Common logarithmic form domain	exact by bounded form perturbation
Central prime-free Weil positivity	Arb-certified
Sharp endpoint-row Schur bound	analytic monotonicity plus Arb scalar enclosure
Degree-four shift expansion and remainder	analytic plus Arb enclosure
All 512 generator boxes	Arb-certified
Infinite Legendre-tail closure	analytic bounds plus certified head
Block-Schur coercivity on $\mathcal{D}_{\log}$	exact (Lemma 7.1)
Independent implementation audit	passed
Strong limit $V_s \rightarrow I$	analytic Hardy-multiplier proof
Closed-form shift-energy identity	analytic proof
Strict contractivity	analytic consequence of generator positivity and compactness
Shift-radial Gram factor	exact consequence
Strict contractivity for $\omega > \frac{1}{2}$	Suzuki [1, Lemma 4.4], not reproved here
Positivity beyond the first slab	open
New zero-free region or RH	not claimed

References

- [1] M. Suzuki, *A canonical system of differential equations arising from the Riemann zeta-function*, arXiv:1204.1827.
- [2] M. Suzuki, *Aspects of the screw function corresponding to the Riemann zeta-function*, arXiv:2206.03682.
- [3] M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096.
- [4] A. Connes and C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. (N.S.) 27 (2021), no. 4, Paper No. 77; arXiv:2006.13771.

- [5] X. Zhu, *Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law*, arXiv:2608.24827 (v2).
- [6] J.-F. Burnol, *Spacetime causality in the study of the Hankel transform*, arXiv:math/0509619.
- [7] J. C. Lagarias, *On a positivity property of the Riemann ξ -function*, Acta Arith. 89 (1999), 217–234; correction, Acta Arith. 116 (2005), 293–294.
- [8] Y. Luchko, *Operational calculus for the general fractional derivatives with the Sonine kernels*, arXiv:2103.00549.
- [9] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green’s functions*, Comm. Math. Phys. 31 (1973), 83–112.
- [10] C. R. Graham and M. Zworski, *Scattering matrix in conformal geometry*, Invent. Math. 152 (2003), 89–118; arXiv:math/0109089.
- [11] NIST Digital Library of Mathematical Functions, Chapter 5: Gamma Function, Sections 5.9, 5.11 and 5.15, <https://dlmf.nist.gov/5>.
- [12] T. Kim, Y. Hong, M. Kim, S. Choi, J. Jang, and M. Kim, *A numerical realization of Suzuki’s Weil-quadratic-form operator: the archimedean spectral law, its universality, and an operator form of Weil’s positivity criterion*, arXiv:2607.24830.
- [13] A. Groskin, *A finite Guinand–Weil dictionary and archimedean tail order for the truncated Weil quadratic form*, arXiv:2607.02828.
- [14] E. Baker, *The shifted zeta string: an unconditional inverse-spectral family and its RH endpoint*, manuscript, August 2026.
- [15] E. Baker, *The margin in the shifted screw-function criterion for zero-free half-planes*, manuscript, August 2026.
- [16] E. Baker, *Detection depth of spectral defects in shifted zeta strings*, manuscript, August 2026.